

Spring 2013: CS 3329/5549 Final Exam

Total Time: 90 minutes

Total Points: 50

Write your name clearly.

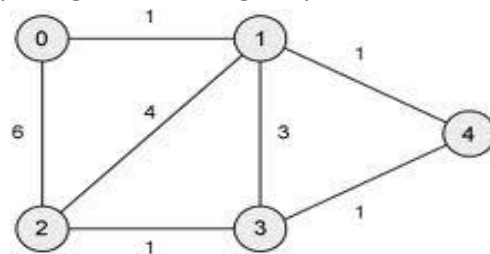
Name: _____

Date: _____

1. **[7]** Describe and analyze quick Sort algorithm.

2. **[5]** What is heapify procedure? What is the run time of min-heapify? Explain

3. **[4+4]** Use Kruskal's algorithm to find a minimal spanning tree for the weighted graph shown below, showing steps. Then, use Prim's algorithm to do the same thing, using the same weighted graph, again showing steps.



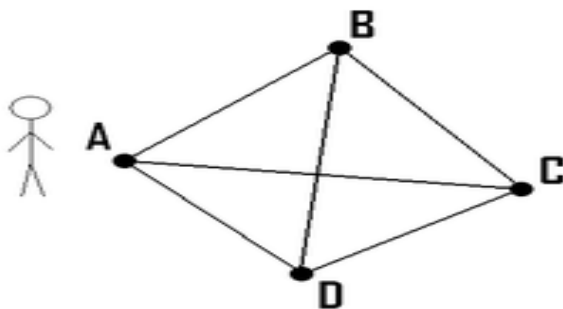
4. [5] Devise an optimization solution of knapsack problem and explain how you would solve this problem using dynamic programming.

5. [3+4] Explain NP completeness. Define set cover problem and explain how it is NP complete.

6. [6] Travelling salesman problem is defined as Finding a path through a weighted graph which starts and ends at the same vertex, includes every other vertex exactly once, and minimizes the total cost of edges.

Given a graph $G = (V, E)$, where V is set of vertices and E is the set of edges. In addition, weights of edges have following properties:

$$e_{ii} = 0; \text{ and } e_{ik} + e_{kj} > e_{ij}$$



A sample graph is shown above

7. [2+2] Given a complete binary tree with node n , find the expression for
- Height of the tree
 - Number of nodes at the lowest level

8. [3] Put the following notation and their description in the right combinations

Θ	Asymptotic Upper Bound
Ω	Asymptotic tight Bound
O	Asymptotic Lower Bound

9. [5] Using the Master Method to give asymptotic bounds for the following recurrences:

a. $T(n) = 9T(n/3) + n$

b. $T(n) = 3T(n/4) + 1$

Master if $T(n) = aT(n/b) + f(n)$ then

Theorem

$$T(n) = \begin{cases} \Theta(n^{\log_b a}) & f(n) = O(n^{\log_b a - \varepsilon}) \\ \Theta(n^{\log_b a} \log n) & f(n) = \Theta(n^{\log_b a}) \\ \Theta(f(n)) & f(n) = \Omega(n^{\log_b a + \varepsilon}) \text{ AND} \\ & af(n/b) < cf(n) \text{ for large } n \end{cases} \begin{matrix} \varepsilon > 0 \\ c < 1 \end{matrix}$$